SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Case Study (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO3	**Assessment:**	Assessment Tool 1 – Case Study: Convolutional Neural Networks
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.***		
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Section / Batch:	AIML	**Date:**	18.8.2026

Code of Conduct
I CH. Durga Prasad (192425305) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: CH. Durga Prasad

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Case Study: Convolutional Neural Networks Duration: 1hr

Case Study

A healthcare organization is planning to develop an AI-based Chest X-Ray Analysis System to assist doctors in detecting pneumonia. The system will use a large dataset of chest X-ray images labeled as Normal and Pneumonia. A CNN model will be trained to automatically learn important visual patterns from the images and classify new X-ray images.
Based on the above case study, answer the following questions:

1. Selection of CNN
Explain why a Convolutional Neural Network (CNN) would be suitable for developing the proposed chest X-ray classification system. Compare its approach to traditional machine learning methods that require manual feature extraction. Discuss how CNNs can automatically learn spatial and visual features from X-ray images.
2. Convolution Layer and Filters
Explain how the convolution layer processes a chest X-ray image. Describe the role of filters/kernels in detecting important patterns such as edges, textures, shapes, and other visual abnormalities. Explain how different filters can learn different features during CNN training.
3. Feature Maps
After applying multiple filters to an input X-ray image, the CNN produces several feature maps. Explain what feature maps represent and how the information captured by feature maps changes as the image passes through deeper CNN layers. Give examples of simple and complex features that may be detected while identifying pneumonia.
4. Pooling Layer
Explain the purpose of the pooling layer in the proposed CNN architecture. Compare Max Pooling and Average Pooling in terms of their working principles and effect on feature maps. Considering the chest X-ray application, explain which pooling technique you would prefer and justify your choice.
5. Fully Connected Layer
After several convolution and pooling operations, the extracted features need to be converted into a final prediction. Explain the role of the Fully Connected (FC) layer in the CNN architecture. How does it use the features extracted from the X-ray to classify the image into Normal or Pneumonia?
6. Parameter Sharing
CNNs use the concept of parameter sharing in convolution layers. Explain how parameter sharing works and why it is important when processing high-resolution medical images. Discuss how parameter sharing reduces the number of trainable parameters compared with a fully connected neural network.
7. Overfitting and Regularization
Suppose the CNN achieves 98% accuracy on the training dataset but only 78% accuracy on unseen X-ray images. Identify the problem occurring in the model and explain why it happens. Discuss suitable regularization techniques, such as Dropout, Data Augmentation, and L1/L2 regularization, that could be used to improve the model's generalization ability.
8. AlexNet Architecture
The development team is considering AlexNet as one of the CNN architectures for the project. Explain the major architectural features of AlexNet, including convolution layers, pooling layers, activation functions, and fully connected layers. Discuss how AlexNet contributed to the development and popularity of deep learning for image classification.
9. AlexNet vs ResNet
The organization is also considering ResNet instead of AlexNet. Compare AlexNet and ResNet based on their architecture, depth, parameter efficiency, training difficulty, and ability to learn complex features. Based on these differences, recommend which architecture would be more appropriate for the chest X-ray classification problem and provide suitable justification.
10. CNN Applications
CNNs are widely used beyond medical image analysis. Explain at least three real-world applications of CNNs, such as facial recognition, autonomous vehicles, object detection, security surveillance, agriculture, or handwriting recognition. For each application, explain what type of image-related problem CNNs solve and how convolution and feature extraction help in solving it.

ANSWERS

1. Selection of CNN

A Convolutional Neural Network (CNN) is highly suitable for developing a chest X-ray classification system because X-ray images contain important spatial and visual patterns such as edges, textures, shapes, and abnormal regions. CNNs are specifically designed to process image data and automatically learn these features.

- **Why CNN is suitable:-**

Traditional machine learning methods such as SVM, Decision Tree, Random Forest, or Logistic Regression generally require the important features to be extracted manually before classification. For example, features such as intensity, texture, edges, or shape may need to be calculated from the X-ray image. The quality of classification therefore depends heavily on the manually selected features.

CNNs overcome this limitation by performing automatic feature extraction. The input chest X-ray is directly provided to the CNN, and convolutional layers learn useful features during training.

- **Working of CNN for X-ray classification**

The CNN processes the image through several stages:

Input X-ray → Convolution → Feature Maps → Pooling → Deeper Convolution Layers → Fully Connected Layer → Classification

In the initial convolution layers, the CNN learns simple features such as:

- Edges
- Lines
- Corners
- Intensity changes
- Basic textures

In deeper layers, these simple features are combined to identify more complex patterns such as:

- Lung structures
- Abnormal textures
- Consolidation
- Opacities
- Patterns associated with pneumonia

➢ **Advantages over traditional machine learning**

Traditional ML	**CNN**
Requires manual feature extraction	Automatically learns features
Feature selection depends on domain knowledge	Features are learned from training images
Image structure may need to be converted into numerical features	Can directly process image data
Difficult to capture complex spatial relationships	Preserves and learns spatial information
Performance depends strongly on feature engineering	Learns hierarchical representations

Therefore, CNNs are appropriate for chest X-ray classification because they can automatically learn hierarchical spatial and visual features from raw medical images and use these features to distinguish between Normal and Pneumonia cases.

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2. Convolution Layer and Filters

The convolution layer is one of the most important components of a CNN. It processes the chest X-ray image using small matrices called filters or kernels.

A filter is moved across different regions of the image. At each position, the filter performs element-wise multiplication with the corresponding image pixels and adds the results. This produces a new value in the output called a feature-map value.

➢ **Role of filters**

Different filters are capable of detecting different types of patterns.

For example:

- One filter may detect horizontal edges.
- Another filter may detect vertical edges.
- Another may detect diagonal edges.
- Some filters may detect textures.
- Deeper filters may detect shapes or abnormal regions.

Initially, the filter weights are usually randomly initialized. During training, the CNN updates these weights using backpropagation and gradient descent so that the filters learn useful features for classification.

➢ **Example in chest X-ray analysis**

Consider a chest X-ray containing a region with abnormal lung opacity.

An early convolution filter may detect the boundary of the opacity. Another filter may identify its texture. Deeper convolution layers can combine these features and recognize more meaningful patterns related to pneumonia.

➢ **Hierarchical feature learning**

The learning process can be represented as:

Pixels → Edges → Textures → Shapes → Lung patterns → Abnormality patterns

Thus, filters in different CNN layers learn different levels of visual information.

➢ **Importance of convolution**

Convolution provides several advantages:

1. It detects important local patterns.
2. It preserves spatial relationships between pixels.
3. Multiple filters can detect multiple types of features.
4. Filters are learned automatically during training.
5. It significantly reduces the number of parameters compared with a fully connected network.

Therefore, the convolution layer acts as the main feature extraction component of the CNN and enables the system to identify useful visual patterns from chest X-ray images.

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3. Feature Maps

A feature map is the output generated when a convolution filter is applied to an input image or a previous feature map.

When multiple filters are applied to a chest X-ray, each filter produces a different feature map. Therefore, if a convolution layer contains 32 filters, it generally produces 32 feature maps.

➢ **What does a feature map represent?**

Each feature map indicates the locations where a particular learned feature is strongly present.

For example:

- One feature map may highlight edges.
- Another may highlight specific textures.
- Another may respond to particular shapes.
- Another may identify abnormal intensity patterns.

The bright or high-value regions in a feature map indicate that the corresponding filter has strongly detected its learned pattern in that region.

- **Feature learning in different layers**

The information becomes more complex as the image moves through the CNN.

Initial layers:

The network learns simple features such as:

- Edges
- Lines
- Corners
- Intensity variations

Middle layers:

These simple features are combined to identify:

- Textures
- Curves
- Shapes
- Lung structures

Deeper layers:

The network learns more complex and meaningful representations such as:

- Abnormal lung patterns
- Opacity regions
- Consolidation-like structures
- Patterns associated with pneumonia

For example, an early feature map might detect a simple boundary inside the lung. A deeper feature map can combine several boundaries and textures to identify a larger abnormal region.

- **Importance in pneumonia detection**

Feature maps allow the CNN to progressively transform the raw X-ray into a meaningful representation.

The process can be represented as:

X-ray pixels → Simple features → Textures and shapes → Complex lung patterns → Pneumonia-related features

Thus, feature maps provide the intermediate representations that allow the CNN to move from low-level visual information to high-level medical image patterns.

4. Pooling Layer

The pooling layer is used after convolution layers to reduce the spatial dimensions of feature maps while retaining important information.

Pooling reduces the size of the feature maps, which decreases computational requirements and helps the network become less sensitive to small changes in the position of features.

Two commonly used pooling techniques are Max Pooling and Average Pooling.

- **Max Pooling:** selects the maximum value from a small region.

For example, using a 2 × 2 pooling window:

2 5

1 3

The maximum value is:

5

Therefore, the output represents the strongest detected feature in that region.

- **Average Pooling:** calculates the average of all values in the pooling region.

For the same region:

2 5

1 3

The average is:

(2 + 5 + 1 + 3) / 4 = 2.75

Thus, Average Pooling provides a smoother representation of the region.

- **Comparison**

Max Pooling	**Average Pooling**
Selects maximum value	Calculates average value
Preserves strongest feature	Preserves overall regional information
Highlights prominent patterns	Produces smoother feature maps
More commonly used in CNNs	Useful when overall information is important

- **Preferred technique for chest X-rays**

For pneumonia classification, Max Pooling can be preferred because abnormal patterns may produce strong activation values in feature maps. Max Pooling retains these strongest activations while reducing the spatial dimensions. For example, if a convolution filter strongly responds to an abnormal lung region, Max Pooling can preserve that strong response.

Therefore, Max Pooling is a suitable choice when the objective is to retain the most prominent learned visual features while reducing computational complexity.

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7. Overfitting and Regularization

Suppose the CNN achieves 98% training accuracy but only 78% accuracy on unseen X-ray images. This indicates that the model is suffering from overfitting.

➢ **What is overfitting?**

Overfitting occurs when a model learns the training data too closely, including noise and dataset-specific patterns, instead of learning general features that work on new images.

In this case:

- Training accuracy = 98%
- Unseen-data accuracy = 78%

The large difference between the two accuracies indicates poor generalization.

➢ **Why does overfitting occur?**

Possible reasons include:

1. The training dataset may be too small.
2. The CNN may contain too many parameters.
3. The training images may not contain enough variation.
4. The model may learn noise or irrelevant image patterns.
5. The model may be trained for too many epochs.

1. Dropout

Dropout randomly disables a percentage of neurons during training.

For example, with a dropout rate of 0.5, approximately half of the selected neurons are temporarily ignored during a training step.

This prevents the network from depending too heavily on particular neurons and encourages it to learn more robust features.

2. Data Augmentation

Data augmentation creates modified versions of existing training images.

For chest X-rays, suitable transformations may include:

- Small rotations
- Horizontal shifting
- Slight scaling
- Cropping
- Intensity adjustments

This increases the effective diversity of the training dataset and helps the CNN handle variations in unseen X-ray images.

Care must be taken that augmentation does not create medically unrealistic images.

3. L1 Regularization

L1 regularization adds a penalty based on the absolute values of model weights.

It encourages some weights to become very small or zero, producing a simpler model and reducing unnecessary complexity.

4. L2 Regularization

L2 regularization adds a penalty based on the squared values of weights.

It discourages excessively large weights and helps the model learn smoother and more stable representations.

➢ **Improving generalization**

A suitable approach would be:

➢ **Data Augmentation + Dropout + L2 Regularization + Proper Validation**

The model should also be evaluated using an independent validation/test dataset rather than relying only on training accuracy.

Therefore, the difference between 98% training accuracy and 78% unseen accuracy clearly indicates overfitting, and regularization techniques can help the CNN learn features that generalize better to new chest X-ray images.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________